EIGSEP Memo 001

The Marjum Pass 2026-07 dataset: observations, events, geometry and conventions

EIGSEP Team

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Abstract

This memo describes the dataset from the EIGSEP deployment at Marjum Pass, Utah, 12–18 July 2026: what was observed, with which instrument configuration, what happened when, and the geometry and angle conventions needed to interpret any of it. It is kept current with our best understanding; it is the reference other memos build on. Every figure and table is generated from the archived data by `make_products.py` in this directory (Appendix C).

1 Summary

- **What:** 5,120 correlator files (11 GB filtered from 87 GB raw), 07-12 13:30 to 07-18 03:08 UTC, 0–250 MHz in 1024 channels of 244 kHz. Two antennas: *box-air*, the science antenna, hung from a highline across the canyon, and *box-gnd*, a static comparison antenna on the ground below.
- **Configuration changed three times in wiring** (phases A, B, C) and **once in integration time** (0.268 s to 0.537 s at 07-15 15:54:59). Only phase C (from 07-15 00:32) has the final wiring: box-gnd on input 0, box-air on input 4.
- **Heights:** box-air at ∼2 m (07-12 to 07-14), ∼30 m (07-15 to 07-16), 87.5 m and finally 91 m (07-17). The deployment ended at 03:05 UTC 07-18 when the pulley plate failed.
- **Transmitter:** the 07-17 beam-mapping transmitter is the comb on channels $\equiv 0 \bmod 8$ (1.953125 MHz spacing). It is first seen at 15:36:22 UTC 07-17, by both antennas. A 1.000 MHz comb of 07-16 is seen only by box-air and matches the comb EMI in the field notes; it is not the transmitter (§6).
- **Geometry:** azimuth rolls box-air about its own boresight, rotating polarization only; elevation moves the boresight within one vertical plane at bearing 37.84°/217.84°. The transmitter is 95.4 m from the 91 m-era antenna, 4.61° from nadir, and is reached at el $\approx$ 184.6° (§7; geometry release `v0004`; memo 002 brackets its positions with held-out checks).

2 The deployment

Site. The instrument was deployed in a canyon at Marjum Pass, Utah (39.2478° N, 113.4026° W). A highline spans the canyon between two ground anchors 420 m apart (Fig. 7); the antenna frame is raised along it on a pulley system tied down to the canyon floor. Site local time is MDT (UTC−6). The field team arrived 07-10 20:40 MDT (07-11 02:40 UTC).

Antennas. Two near-identical dipole antennas, each with its own receiver box:

- **box-air** is the science antenna. It is mounted on a frame raised up the highline, which minimizes coupling to terrain and lets the beam be mapped in situ. It carries motorized azimuth and elevation drives, and the digital control electronics (the "Panda" computer, motor controllers, IMU, LIDAR rangefinder, potentiometer, RF switch and noise source).
- **box-gnd** is a comparison baseline. It sits flat on the ground below, with no motors and no digital control electronics. It is stable, static and free of that electronics' self-generated RFI, but it couples strongly to terrain.

Only box-air's signal path is switched. Its RF switch cycles among the antenna (`RFANT`), a noise source (`RFNON`), an ambient load (`RFAMB`), a terminated noise-off state (`RFNOFF`) and a spare port (`RFSP1`), and routes VNA standards (`VNAO`, `VNAS`, `VNAL`, `VNAANT`) for reflection measurements. Box-air moves by a factor 2.3 between `RFNON` and `RFAMB` dwells while box-gnd moves by less than 2% (`INDEX.md` §Calibration constraints), so box-gnd is always on its antenna.

Transmitter. A beam-mapping transmitter sits on the canyon floor below the highline (position in Table 6). The field notes' "Tx setup, final configuration" on 07-17 lists two LNAs, 13 dB attenuation and an SLP-200+ low-pass filter (p. 181). It radiates a comb whose teeth fall on correlator channels $\equiv 0 \bmod 8$ (§6). The transmitter antenna is a pair of crossed dipoles ("arms") that radiate alternate teeth. **Arm 0** radiates the teeth on channels $\equiv 0 \bmod 16$, at $n \times 3.90625\,\mathrm{MHz}$ (e.g. 156.25, 160.16 MHz); **arm 1**, polarized at 90° to arm 0, radiates the teeth on channels $\equiv 8 \bmod 16$, at $(n+\frac{1}{2}) \times 3.90625\,\mathrm{MHz}$ (e.g. 158.20 MHz). This is the convention of `eigsep_data.beam_mapping` (`tx_teeth.tooth_arms`: arm = $(c \,//\, 8) \bmod 2$ for channel c).

Arm 0 is the dipole driven by receiver chain rx6, with its axis at 166°/346° true (NNW–SSE); arm 1 is the rx1 dipole, at 76°/256° (Fig. 1; the bearings are measured in memo 002 at geometry v0004). A screenshot of the live correlator display taken with only chain rx6 powered carries its capture time, 15:56:29 UTC on 07-17. The recorded data place it: from 15:54:53 to 15:56:22 only arm 0 radiated, and arm 1 came back on at 15:56:22.6, so the display shows the end of that interval (a few seconds' display latency or clock offset). Over its 152 integrations in `corr_20260717_155704Z.h5` every tooth is on a channel $\equiv 0 \bmod 16$; the arm-1 teeth are at least -40 dB below the arm-0 teeth on box-air and -33 dB on box-gnd. In the field the diagonal pair of feed quadrants wired to rx6 was marked "6" (IMG_2204, looking down on the transmitter at 10:06 MDT on 07-17). The pair's axis is measured from the feed's corners in IMG_2203 at 165.5°±1° true (mod 180), 1.2° from IMG_2204's theodolite-app reading, whose north reference is not recorded (memo 002 §9). Christian's analysis of the 07-17 raster agrees that the two sets of teeth are orthogonal polarizations: they couple to box-air in antiphase as its azimuth steps, the arm-1 teeth nulling at motor azimuth $-130°$ where the arm-0 teeth peak, and the reverse near $-45°$ (`data-analysis/notebooks/christian/beam_modulation`). The reference beam fit (`derived/beam/empirical_raster_v0012`) puts arm 0's field at $\alpha \approx 96°$, a field along bearing 84°/264°, 83° from the measured rx6 axis and close to rx1. The arm ratio alone, without a beam model, does the same. With the arms held at their photographed bearings, memo 003 (§Absolute azimuth) finds that only the right-handed azimuth fits well, and only if arm 0 lies along the axis photographed as rx1: one of the labels in the chain from the display's "rx6" to the feed quadrants is probably swapped.

3 Observation parameters

Table 1 lists the correlator configuration, read from the file headers. Each file holds 240 consecutive spectra of every recorded auto-correlation and cross-correlation, plus `header/` and `metadata/` groups carrying per-spectrum timestamps and the telemetry of the switch, motors, IMU, LIDAR, potentiometer and temperature controllers. The filter that produced the archived set (`data/filter_corr_keys.py`) kept only the correlator inputs that carried antennas in each phase (Table 3) and all of `header/` and `metadata/`.

Table 1: Correlator configuration (from the file headers).

ADC sample rate	500 MHz (Nyquist band 0–250 MHz)
Channels	1024 × 244.141 kHz; channel k centred at $k \times 250/1024$ MHz
Integration	corr_acc_len 2^{26} = 0.268 s until 07-15 15:54:59 UTC, then 2^{27} = 0.537 s (1,597 files); 2^{28} in the 8 SNAP-flip files
Spectra per file	240 (64.4 s or 128.8 s per file); 7 files are short
Data type	int32; autos `data/<i>` (time, 1024); crosses `data/<ij>` (time, 1024, 2) = (re, im)
ADC gain / FFT shift / corr scalar	4 / 351 / 512
Firmware	`eigsep_fengine_1g_v2_4_2026-06-23_2052.fpg`
SNAP board	C000122 (flight); C000069 only in the 8 test-flip files
Files / volume	5120 files, 11 GB filtered (87 GB raw, off-line)
Span	07-12 13:30 → 07-18 03:08:49 UTC (last sample)

Time. All times in this memo are UTC unless marked MDT. Two clocks exist and neither is always right:

- `header/times` is the per-spectrum timestamp. It is authoritative where valid, but in **646 files it is not**: it disagrees with the filename by more than an hour, or is non-monotonic within the file. These are 599 files from 07-13 05:05–15:57 with the clock ∼54 days in the past, the 8 SNAP test-flip files, and ∼40 scattered others. From 07-15 16:00 onward only the test-flip files are affected; the 07-17/18 beam-scan window is clean.
- The filename is the file's *close* time. It identifies and orders files, but it trails the last sample by a median of ∼630 s on 07-12 and 07-15 and by ∼1 s on other days. Buffered writes also give 572 files the same close time as another file (hence the `-1`, `-2` filename suffixes), e.g. ten files stamped 07-15 15:56:22–24. Where the header clock is bad, this memo places a file at its close time minus half its span.

The `curation/pointing_table.parquet` column `hdr_time_bad` marks the 646 files; it and this memo's independent count agree.

4 Configuration history

Wiring phases. The antennas were moved between correlator inputs twice (Table 3). Each file's `header/wiring` records the antenna-to-SNAP-input mapping, but it holds only two configurations over the whole campaign (Table 2), and before 07-15 03:19 it names inputs that carry no antenna. Fig. 2 shows every recorded input: an antenna's spectrum carries sky, FM and RFI lines, while an input with nothing connected shows only the smooth receiver bandpass. From 07-13 05:00 (the field notes' first hillside observing) input 3 carries an antenna, and input 4 from ∼19:00 07-13; inputs 0 and 2, which the header calls box-air and box-gnd, stay flat until the B→C move. After that move inputs 0 and 4 are live at 01:17, about two hours before the header changes to box-gnd on 0 and box-air on 4. The phase boundaries are therefore taken from the data (liveness, byte-identical mux copies, power levels); from 07-15 03:19 the header and the data agree. In phase B input 4 is box-air and input 3 box-gnd: switching the Panda off on 07-14 removes box-air's excess from input 4 and barely changes input 3 (memo 007 §Inputs). Which antenna was on which input in phase A is open (§9).

Table 2: The antenna-to-input mapping recorded in `header/wiring` (SNAP test flips excluded).

first file	last file	header `wiring` (antenna: input)	files
20260712_133146	20260715_030236	box-air: 0, box-gnd: 2	2835
20260715_031915	20260718_032248	box-gnd: 0, box-air: 4	2277

Table 3: Wiring phases. Times are file close times; hours sum the file spans.

phase	first file (close, UTC)	last file	files	hours	live inputs
A	20260712_133146	20260714_040243	1588	28.4	3 (+4); 0 and 2 possibly early (unconfirmed). No 34 cross.
B	20260714_041043	20260715_003113	1111	19.9	3 = box-gnd, 4 = box-air (memo 007); mux copy 4→5 gives the 35 cross.
C	20260715_003217	20260718_032248	2421	71.9	0 = box-gnd, 4 = box-air; cross 04. Mux copies 0→1, 4→5 from 03:29.

In phase C, the ADC multiplexer copies input 0 to 1 and 4 to 5 from 07-15 03:29; those copies are byte-identical duplicates and were dropped by the filter. Phase A has no valid cross between the two antennas. 07-14 (most of phase B) was a deliberate shakedown day: the system was being exercised and debugged, so treat it as engineering data.

Heights. Box-air's height history comes from the field notes; the LIDAR reports in only 244 files and saturates at 250 m. Table 4 lists the eras. The three numbers quoted for the final height measure different things and must not be averaged (`curation/height_references.json`). In the field on 07-12, the pulley plate was observed to deflect from the unloaded height of 97 m by ∼2 m under a 160 lb load (field notes p. 176).

Table 4: Box-air height eras and what each height number is referenced to.

era	from (UTC)	height and datum
~2 m	07-12	On portable workbench near box-gnd.
~30 m	07-15	"Lift to ~30 m completed on 7/15" (p. 179). No datum.
87.5 m	07-17 02:30	"Measured from tether anchor" (p. 180), the only height with a stated datum. The anchor's own position is not recorded.
91 m	07-17 18:50	Field note, no datum (p. 181). The same era is 94.1 m above the DEM surface at the geometry-`v0004` antenna position (Table 6), and the LIDAR reads 92.3 m, a slant range ~2.3° off vertical to whatever ground it hit.

Integration time. `corr_acc_len` doubled from 2^{26} to 2^{27} at 07-15 15:54:59 (file `corr_20260715_155459Z.h5`), doubling the integration from 0.268 to 0.537 s and the file span from 64 to 129 s. Power scales with `corr_acc_len`, so the figures here, which plot correlator counts, show a factor-2 (3 dB) step at this point; normalize by it or split any analysis here. The doubling delivered the expected $\sqrt{2}$ reduction in radiometer noise. It also made the int32 auto-correlation accumulators wrap more often on the brightest channels: files with at least one wrapped sample rose from 5.4% to 25.6%, and to 47.6% in the 91 m beam-scan era, with 155.8 MHz the worst channel on every input. A wrapped sample reads negative and is repaired exactly by adding 2^{32}; repair rather than mask (`experiments/acclen_wrap_scan.jsonl`, `curation/overflow_channels.jsonl`). The accumulator stores power modulo 2^{32}, so a true value between 2^{31} and 2^{32} reads negative, but one above 2^{32} wraps twice and reads as a small *positive* number that looks valid. Such a sample shows only as a drop of $\sim 2^{32}$ against the neighbouring integrations. In the 227 files from 07-17 18:50 onward, 3,107 samples of inputs 0 and 4 wrap once and 37 more wrap twice: each sits below both neighbours by more than 2^{31}, and adding 2^{32} again makes it continuous with them. They occur in the brightest RFI channels (e.g. 133.5, 135.7, 155.8 MHz). A channel that stayed above 2^{32} for a run of integrations would show no such drop and could not be repaired, so treat any channel whose repaired power reaches $\sim 2^{32}$ as saturated.

Calibration cadence. The RF switch daemon was not running for the first 1,842 files (all of phase A and early phase B), so they carry no switch record. The first VNA sweep was at 07-13 03:47. From 07-17 04:12 the cadence is regular: an `RFNON`+`RFAMB` pair every ~13 min and a full open/short/load sweep about every fourth cycle, until 07-18 03:00. Raw VNA sweeps are in `data/s11/`; the switch states in the correlator stream are timing markers only.

5 Timeline

Figure 3 is the whole campaign at a glance. The full event list, with sources, is in Appendix A.

07-12 to 07-14: ground tests, box-air at ~2 m. First data 07-12 13:31. RFI shakedown on 07-13: laptop RFI at 145–160 MHz, fan RFI near 150 MHz, CU battery work. On 07-14 (00:00 UTC) the Panda box's power connector and RJ45 ports were found to leak EMI and were gasketed; the LIDAR and external IMU were unplugged as RFI sources; a wide passband on both antennas was traced to swapped front-end cables and corrected. Wiring phase A→B at 07-14 04:10:43. Inputs 4/5 unplugged 18:33–20:33 and the phase-B mux copy off 21:22–23:31.

07-15: final wiring, lift to ~30 m. Phase B→C at 00:32:17; the next 45 minutes are all-zero while the antennas were moved. The mux copies came online at 03:29 and were then tuned in seven windows between 05:39 and 15:56. The integration time doubled at 15:54:59. Box-air outage 18:55–21:30. A thunderstorm began at ~00:00 07-16.

07-16: comb EMI, troubleshooting, raise to 87.5 m. At 01:16:57 a 1.000 MHz comb appeared on box-air (§6); the field notes record a comb of spikes at 01:00–01:30. The box was lowered; the comb survived a CU power cycle and disappeared when the Panda was power-cycled, which the data place between 16:49:03 and 16:51:12. The elevation drive's helical-gear set screw was replaced. Box-air outages at 17:32 and 18:42–23:27. Raised to 87.5 m by 07-17 02:30.

07-17: final height and the beam scan. Three parallax photos of the antenna at 87.5 m (15:05–15:36). The transmitter comb first appears at 15:36 and was then switched on and off, and its drive attenuation changed, repeatedly until 18:14 to test for transmitter LNA noise outside the comb and to try attenuation settings (§6); it was off 17:04–17:55. The final configuration, used for the raster, holds from 18:14:47 to the end of the data. Transmitter on/off test at ∼18:44; lift to the final 91 m at ∼18:50. The transmitter beam-mapping raster ran 20:26–21:28: the elevation axis turned through continuous full revolutions while the commanded azimuth stepped 5.0° once per revolution, 52 times, from 190.6° through 360° to 90.6° (memo 003; the potentiometer, which slipped, reads ∼6° steps). Positioning photos 20:42–21:10 (34 frames). The 7U sky battery failed at ∼22:00; box-air was out 21:44–23:44. The change in box-air's spectrum on its return at 23:44 is the transmitter comb reappearing, not a change in the receiver: the ambient-load power moved only 4–12% across the gap.

07-18: end. LIDAR sweep from 01:37, beam map resumed at 01:53, and the elevation drive failed at 01:41:38, parked at el $\approx$ 184°, facing the transmitter (memo 003; the field notes give 01:55). The field notes suspected LIDAR-linked RFI every 15 min; no periodicity near 15 min stands out in the data, per sample or per file, on either antenna. At 03:05 the bottom pulley plate split. The last valid sample is 03:08:49; the last three files were flush-written up to 931 s late, so their filenames overstate the post-failure data.

6 Combs, and which one is the transmitter

Two combs of narrow tones appear in the data. One, on channels $\equiv 0 \bmod 8$ from 07-17, is the beam-mapping transmitter. The other, at 1.000 MHz spacing on 07-16, is electromagnetic interference from box-air's own electronics and must not be confused with it. This section measures both across the whole campaign.

Method. For each file and input, take the median spectrum over the file's integrations. For each comb family, define *tooth* channels (nearest to the tone frequencies) and *gap* channels (nearest to the half-spacing offsets) in 110–230 MHz, drop channels shared with another family's teeth, and form the contrast $10\log_{10}$(median tooth power / median gap power). With no comb present the contrast is ≈ 0 dB. Three families are tested: 8 channels (1.953125 MHz, teeth at channels $\equiv 0 \bmod 8$), 1.000 MHz (teeth at integer MHz, which walk across the channel grid), and 1.25 MHz (the spacing in the field notes).

The 8-channel comb is the transmitter. It switches on at **07-17 15:36, in both antennas at once**, toggles on and off together until 17:04 (the tests below), is off 17:04–17:55, and then runs to the end of the data (on box-gnd to 02:59; on box-air to 21:27 and again from 23:43, the gap being box-air's outage). Box-gnd sees it at up to ∼27 dB contrast: it sits on the ground below the highline, as the transmitter does. On box-air the tooth-to-gap ratio swings with the elevation sweeps of the raster, which is what a beam pattern does to an external source. Consecutive teeth alternate strong and weak (Fig. 5b), because alternate teeth are radiated by orthogonal transmitter polarizations (below). For transmitter beam mapping use the tooth channels, preferably 170–200 MHz where the transmitter dominates; the mid-gap channels ($\equiv 4 \bmod 8$) carry the sky and terrain under it.

Alternate teeth are orthogonal polarizations. Teeth on channels $\equiv 0 \bmod 16$ come from arm 0 (rx6, axis 166°), and teeth on channels $\equiv 8 \bmod 16$ from arm 1 (rx1, axis 76°) (§2, Fig. 1). Each antenna therefore sees two interleaved combs whose ratio is set by how its own dipole lines up with each arm and by how hard each arm is driven. The arms were driven at different levels (§6): in the final configuration box-gnd sees arm 0 about 17 dB below arm 1, whereas at 16:05–16:14, with the arms set equal, it saw them equally. The arm ratio is therefore not a clean polarization measurement unless the drive levels are known. Beam mapping must model the two arms separately (`PolarizationBeamMapper`).

The 1.000 MHz comb is box-air self-EMI. It is present **only on box-air**, from 07-16 01:16:57 to 16:49:03, switching fully on and fully off within one file at each end, at a steady ∼10 dB. Box-gnd, on the ground near the transmitter, never shows it (Fig. 5a), which an external transmitter could not produce. Its onset matches the field notes' comb of spikes at 01:00–01:30, and its end matches their account that the comb survived a CU power cycle but vanished when the Panda was power-cycled. It should be masked on box-air over that window.

Transmitter tests on 07-17. Between 15:36 and 18:14 on 07-17 the transmitter was switched on and off and its drive attenuation was varied, to find a setting that radiated no LNA noise between the teeth. Box-gnd does not move, so any change in the tooth power it sees is a change in the transmitter. Fig. 6a and Table 5 give the tooth power of each arm (above the neighbouring gaps, 150–200 MHz) in each steady segment, relative to arm 1 in the final configuration. The arms were set independently. At first arm 1 was at +8.5 dB and arm 0 at +1.3 dB; at 15:44 both dropped, to +0.6 and −9.9 dB; 15:54–16:14 alternated between settings, mostly with the arms equal (16:05–16:14 steadily so); from 16:22 to 18:13 came single-arm and weaker settings, with arm 1 as low as −21 dB and arm 0 often absent. From 18:14:47 to the end of the data the drive is steady: arm 1 at 0 dB, arm 0 at −16.9 dB, drifting to −15.5 dB after 07-18 00:00.

Table 5: Steady transmitter drive segments seen by box-gnd (at least 20 s; segments within 1.5 dB and 2 min merged). Times are MDT; the file is the one being written at the segment's start (Appendix A). The tests of 15:36–18:13 are one row; their settings are in Fig. 6a. Tooth power per arm above the neighbouring gaps, 150–200 MHz, relative to arm 1 in the final configuration; "absent" is below −30 dB.

start (MDT)	end	file at start	seconds	arm 1 [dB]	arm 0 [dB]
07-17 09:36:22	12:13:13	`corr_20260717_153744Z`	9410	various	various
07-17 12:14:47	15:28:32	`corr_20260717_181639Z`	11625	+0.0	-16.9
07-17 15:42:11	18:00:10	`corr_20260717_214419Z`	8279	-0.0	-16.6
07-17 18:00:11	21:00:18	`corr_20260718_000145Z`	10807	+0.0	-15.5

The comb's spectral shape also changed: in the final configuration the arm-1 teeth at 200–240 MHz are 9.9 dB below those at 150–200 MHz on box-gnd, against 3–8 dB in the earlier steady settings, a steeper roll-off consistent with the SLP-200+ low-pass the field notes list for the final configuration.

The on/off toggles test the noise directly (Fig. 6b, c). The transmitter's state is taken per integration from box-gnd's tooth/gap contrast; for each toggle with at least 10 steady integrations on both sides, the mean spectrum with the transmitter off is subtracted from the mean with it on, using only channels at least two from any tooth, and the median over 40–240 MHz taken. The off-comb power itself is 5.5×10^5 correlator counts. At the first toggles (15:36–15:38) switching the transmitter on raised it by 3.2×10^5 counts on box-gnd and 2.5×10^5 on box-air: the transmitter's LNA noise. The excess fell to 5.3×10^4 (box-gnd) at 15:39–15:43 and 1.1×10^4 at 16:14. From 16:36 onward it is at most 2.0×10^3, within the $\pm 5.0 \times 10^3$ spread of the same statistic across adjacent transmitter-off stretches. The configuration used for the raster therefore adds no detectable noise outside the comb. Fig. 6c shows the last switch-on in full: between the teeth, on − off follows its own 1σ noise, and 164 (box-gnd) and 227 (box-air) of the 513 off-comb channels from 0 to 250 MHz come out negative, at that noise level. One toggle, 15:50:34, shows the off-comb power *falling* by 1.6×10^5 counts on box-gnd when the transmitter comes on; it is not understood. Nor is the drop in off-comb noise from 3.3×10^5 to 5.3×10^4 counts between the toggles at 15:38:30 and 15:39:09: it happened in every band above 110 MHz while the tooth power in every band stayed the same, so it was not an attenuation change.

No 1.25 MHz comb is detected. The field notes' "every 1.25 MHz" is most likely the 1.000 MHz comb, whose onset is inside the noted window. The only positive 1.25 MHz contrast is a ∼1.5 dB excursion on input 3 on 07-13 21:03–23:03, which this memo does not interpret.

7 Geometry and conventions

Mount. The azimuth rotation is applied first, about the antenna's mast, and the elevation rotation second, tipping the box and antenna about an axis along the highline: $R = R_{\text{highline}}(\text{el})\, R_{\text{mast}}(\text{az})$. The mast is collinear with the boresight, so **azimuth does not steer the beam; it rolls the antenna about its boresight and**

Table 6: Reference positions on the UTM raster grid of DEM `v0002` (EPSG:6341 minus the raster origin, metres). Antenna and transmitter are geometry release `v0004` (memo 002), a point fit whose bounds are hard brackets from held-out checks, not sigmas. Anchor and tiedown horizontal positions are GPS (receiver accuracy unrecorded), placed with no horizontal survey offset; their U is the DEM surface.

point	E, N, U [m]	source
Antenna (box-air), 91 m era	(1656.2, 2031.8, 1778.6)	geometry release v0004
Transmitter	(1652.2, 2025.3, 1683.5)	geometry release v0004
Highline east anchor	(1803.0, 1903.2, 1848.2)	GPS; U from DEM
Highline west anchor	(1478.4, 2169.5, 1867.3)	GPS; U from DEM
Pulley-plate tiedown	(1662.0, 2034.1, 1684.8)	GPS; U from DEM
Antenna height	94.1 m above DEM at antenna (E, N)	see Table 4
TX from antenna	95.4 m; true bearing 210.4°; 4.61° from nadir	derived

changes only the polarization angle. The boresight is moved by elevation alone, within the vertical plane perpendicular to the highline, at bearing 37.84°/217.84° (Fig. 7). The transmitter polarization through the raster rules out any alt-az variant (memo 003).

Elevation. el is the boresight's angle from zenith, by the right-hand rule about the highline pointing west-north-west (bearing 307.84°): el = 0 is zenith, +90 is horizontal toward bearing 37.84°, 180 is nadir. The transmitter is reached at el ≈ 184.6°, not at nadir; it lies 0.6° out of the boresight plane. No uncertainty is attached here: memo 002 bounds the two positions by hard brackets, not sigmas. The LIDAR is mounted 90° from the boresight, pointing horizontally toward 37.84° at el = 0, so it points at nadir when el = 90. The IMU elevation reads about −2.3° at zenith: LIDAR ground returns cluster at IMU el ≈ 87.7°.

Azimuth. Azimuth is by the right-hand rule about the mast (up, at el = 0), so increasing azimuth turns the dipole counter-clockwise seen from above and decreases its compass bearing. At potentiometer azimuth 0 the dipole arm lies along the highline (bearing 127.84°/307.84°), from the GPS anchor bearing and the hardware layout (±0.6° for ±3 m anchor errors). That is the definition; the table's azimuth, which inherits the potentiometer's zero, is 16° from it on the 07-17 raster (memo 003). We write Δaz for the rotation from this reference position (Fig. 7c); it is a rotation of the dipole about the boresight, not a compass bearing. A terrain fit of the LIDAR ranges at the `v0004` antenna, with these true bearings applied as grid bearings, puts it at 36.27°, 1.57°±0.75° from the GPS value: a 2.1σ tension that is not yet explained (memo 002). The pointing table's `az_deg` is the *commanded* azimuth since v2.0 (motor counts, with one potentiometer offset per anchor segment): on 07-17 the potentiometer slipped by ∼27° while the platform followed the motor, and the commanded azimuth fits the beam data much better (held-out residual 0.071 against 0.197). The potentiometer-fused value survives as `az_fused_deg`. The two drives never move at the same time and are locked when idle.

Coordinates and time. Positions are E, N, U in metres on the UTM raster grid of `derived/dem/v0002`: EPSG:6341 easting and northing minus the southwest pixel-centre origin (291000.25, 4345000.25) m, with U on the DEM's vertical datum, which has not been independently verified (memo 002). Raster axes are UTM grid axes, so a grid bearing exceeds the true bearing by 1.53° here. Bearings quoted in this memo are compass degrees clockwise from true north. Times are UTC; MDT = UTC−6 h.

8 Selecting data

Use `eigsep_data.select_files` (CLI `eigsep-select-files`), which applies the masks in Appendix B and reports what it drops. Most science wants phase C only; state the phase, inputs, masks and frequency range with any result. The default mask `boxair-emi-1mhz` removes the 1.000 MHz box-air self-EMI of 07-16 01:17–16:49 from box-air; memo 006 gives the comb masks and the corrected comb names.

9 Open questions

- Which antenna was on input 3 and which on input 4 in phase A (phase B: memo 007). The header `wiring` names inputs 0 and 2 there, which carry no antenna (Fig. 2).
- Why the off-comb power fell when the transmitter came on at 15:50:34 on 07-17, and what cut the transmitter's off-comb noise between 15:38:30 and 15:39:09 without changing its comb.
- The absolute transmitter drive per arm in the final configuration; the data give only the ratio as seen by box-gnd.
- Which feed quadrant pair the display's "rx6" output drives: the transmitter data put arm 0 along the axis photographed as rx1 (memo 003, Absolute azimuth).
- The tether anchor's position, needed to convert the 87.5 m and 91 m field heights to a datum.

Revision history

2026-10-10 The transmitter-feed axes are now true bearings at geometry v0004 (rx6 165.5°); the earlier 167.3° was a bearing on the UTM grid.

2026-10-08 Positions now come from geometry release `v0004` (memo 002), with no uncertainty attached; the transmitter is 95.4 m away and 4.61° from nadir (93.3 m and 4.23° in release `v0001`). The raster grid is the UTM grid, 1.53° from true north, not a tangent grid aligned with true north.

2026-09-30 First version. Establishes from the data that the 07-17 comb on channels $\equiv 0 \bmod 8$ is the transmitter, seen by both antennas, and that the 07-16 1.000 MHz comb is box-air self-RFI. Same day, after review: adds the header wiring record and the evidence that it is stale before 07-15 03:19, the transmitter's alternating-arm polarization, and the 07-17 toggle test of transmitter LNA noise. Then: the 07-17 drive-attenuation record per arm, the final configuration from 18:14:47, and the change in comb shape; withdraws the reading of the arm ratio as box-gnd's polarization. Later: the transmitter arms' bearings from memo 002 (rx6 axis 167°, rx1 77°), replacing "roughly north–south", which rested on a phone compass.

2026-10-01 Pointing details moved to memo 003: the raster's exact azimuth steps (from the motor, not the slipping potentiometer), the elevation failure at 01:41:38, and the arm-exchange question: memo 003 finds a swapped arm label more likely than a wrong azimuth handedness.

2026-10-03 Phase B's input mapping settled (input 3 box-gnd, input 4 box-air) by the Panda power-off tests of memo 007.

A Event list

Generated from `marjum-2026-07/events.jsonl`; first sentence of each note; sources are field-notes pages (`fieldnotes:p`), the data README (`readme`), the per-file scan (`scan:*`, `csv`) and inference from file times (`derived`). Times are MDT (UTC−6 h). The file column names the data file being written when the event started: the first file whose name (its UTC close time) is at or after the start. A dagger marks events in a recording gap, where that file began after the event; a dash, events more than an hour from any data.

MDT	file at start	category	event (source)
07-10 20:40	—	logistics	Party arrives on site (20:40 MDT 07-10). (fieldnotes:p176)
07-11 00:00–18:00	—	mechanical	Drone flights (twice, because the fishing line snapped) and raising the pulley plate. (fieldnotes:p176)
07-12 00:00–07-13 00:00	—	mechanical	Loaded to 160 lb. (fieldnotes:p176)
07-12 07:31	`corr_20260712_133146Z`	data-start	First correlator file of the campaign (Phase A). (derived)

MDT	file at start	category	event (source)
07-12 14:34	corr_20260712_203438Z	data-start	Dense recording resumes after ∼7 h gap from morning first file. (derived)
07-12 15:15–15:16	corr_20260712_211540Z	config-change	Brief (∼79s) SNAP swap: flight config (C000122) → scratch (C000069) → back. (scan:snap-test-flip)
07-12 23:00	corr_20260713_050509Z[†]	config-change	First hillside observing (∼23:00 MDT 07-12). (fieldnotes:p177)
07-13 00:00–07-14 00:00	corr_20260713_060059Z	rfi	Christian's laptop emits 145–160 MHz with 2 MHz spacing during hillside checks. (fieldnotes:p177)
07-13 00:00–07-14 00:00	corr_20260713_060059Z	observation	Hillside diagnostics. (fieldnotes:p177)
07-13 16:00–18:00	corr_20260713_220100Z	power	∼2 h ground-test run starting ∼16:00 MDT. (fieldnotes:p177)
07-13 18:22–18:38	corr_20260714_003801Z[†]	data-gap	Recording gap of 15.7 min between files. (scan:gap)
07-13 20:07–20:09	corr_20260714_020757Z	config-change	Brief (∼86s) SNAP swap: flight config (C000122) → scratch (C000069) → back. (scan:snap-test-flip)
07-13 22:10	corr_20260714_041043Z	phase-transition	PHASE A → B. (csv+readme)
07-14 00:00–07-15 00:00	corr_20260714_060049Z	config-change	Panda EMI shielding: gasket around power connector + RJ45 ports; BLP-5 caps + BLP-6 on Panda power input. (fieldnotes:p177)
07-14 00:00–07-15 00:00	corr_20260714_060049Z	config-change	Wide passband observed on sky output for both dipole + blade → suspected cable swap (blue↔silver). (fieldnotes:p177)
07-14 00:00–07-15 00:00	corr_20260714_060049Z	observation	Shakedown day: the system was deliberately exercised and debugged ahead of science observing (EMI work, front cable swap, LIDAR/IMU unplugged, A→B wiring change). (aaron:2026-09-20)
07-14 00:00–07-15 00:00	corr_20260714_060049Z	logistics	Satellite contact established. (fieldnotes:p178)
07-14 12:33–14:33	corr_20260714_183323Z	outage	Phase B: inputs 4/5 unplugged. (readme)
07-14 15:22–17:31	corr_20260714_212233Z	config-change	Phase B mux OFF. (readme)
07-14 18:00	corr_20260715_002550Z[†]	power	∼18:00 MDT 07-14: swapped two 12 V ground batteries at receiver (fully discharged). (fieldnotes:p178)
07-14 18:25	corr_20260715_002550Z	config-change	∼18:25 MDT 07-14: RF switch connected, LIDAR not (cross-contamination). (fieldnotes:p179)
07-14 18:32	corr_20260715_003217Z	phase-transition	PHASE B → C. (csv+readme)
07-14 19:17–21:29	corr_20260715_011723Z	config-change	Phase C: 0/4 live without mux copies yet. (readme)
07-14 21:29	corr_20260715_032919Z	config-change	Phase C: mux copies 0→1 and 4→5 come online. (readme)
07-14 22:43	corr_20260715_044826Z[†]	data-gap	Corrupt source file skipped by filter. (readme)
07-14 23:39–07-15 00:14	corr_20260715_053958Z	config-change	ADC-mux tuning window: 6 rapid changes to mux_0to1/mux_4to5 (34 min). (scan:mux-tuning)
07-15 00:00–07-16 00:00	corr_20260715_060129Z[†]	mechanical	Lift to ∼30 m completed on 7/15; left overnight. (fieldnotes:p179)
07-15 00:47–01:16	corr_20260715_064739Z	config-change	ADC-mux tuning window: 8 rapid changes to mux_0to1/mux_4to5 (29 min). (scan:mux-tuning)
07-15 02:04–02:13	corr_20260715_080457Z	config-change	ADC-mux tuning window: 6 rapid changes to mux_0to1/mux_4to5 (9 min). (scan:mux-tuning)
07-15 02:51–04:53	corr_20260715_085112Z	config-change	ADC-mux tuning window: 16 rapid changes to mux_0to1/mux_4to5 (122 min). (scan:mux-tuning)
07-15 05:48–06:18	corr_20260715_114814Z	config-change	ADC-mux tuning window: 8 rapid changes to mux_0to1/mux_4to5 (30 min). (scan:mux-tuning)
07-15 07:34–08:50	corr_20260715_133439Z	config-change	ADC-mux tuning window: 15 rapid changes to mux_0to1/mux_4to5 (76 min). (scan:mux-tuning)

MDT	file at start	category	event (source)
07-15 09:09–09:25	`corr_20260715_152502Z`†	data-gap	Recording gap of 15.6 min between files. (scan:gap)
07-15 09:25–09:56	`corr_20260715_152502Z`	config-change	ADC-mux tuning window: 6 rapid changes to mux_0to1/mux_4to5 (31 min). (scan:mux-tuning)
07-15 09:54	`corr_20260715_155459Z`	config-change	Correlator accumulator length changed: 67108864 → 134217728. (scan:corr-acc-len)
07-15 12:55–15:30	`corr_20260715_185550Z`	outage	Phase C box-air (input 4) outage. (readme)
07-15 15:31	—	data-gap	Corrupt source file skipped by filter. (readme)
07-15 18:00	`corr_20260716_000042Z`	weather	Thunderstorm begins ∼18:00 MDT 07-15. (fieldnotes:p179)
07-15 19:00–19:30	`corr_20260716_010050Z`	rfi	RF resonance event: combs of spikes every 1.25 MHz, lower spikes at half that. (fieldnotes:p179)
07-16 00:00–07-17 00:00	`corr_20260716_060134Z`	config-change	Lowered box; comb EMI still present after CU cycle. (fieldnotes:p180)
07-16 00:00–07-17 00:00	`corr_20260716_060134Z`	mechanical	EL motor fully slipping: helical-gear set screw not seated. (fieldnotes:p180)
07-16 00:00–07-17 00:00	`corr_20260716_060134Z`	mechanical	Peltier and mechanical testing before the re-raise: EL stalls at N 55 deg (up) and S 110 deg (down); az contacts the frame with the blade on the east side. (fieldnotes:p180)
07-16 11:32	`corr_20260716_173215Z`	outage	Phase C box-air (input 4) outage. (readme)
07-16 12:42–17:27	`corr_20260716_184257Z`	outage	Phase C box-air (input 4) outage. (readme)
07-16 20:30	`corr_20260717_023204Z`	mechanical	Raised to ∼87.5 m (measured from tether anchor) by ∼20:30 MDT 07-16. (fieldnotes:p180)
07-16 21:24–21:27	`corr_20260717_032454Z`	config-change	Brief (∼129s) SNAP swap: flight config (C000122) → scratch (C000069) → back. (scan:snap-test-flip)
07-16 22:00	`corr_20260717_040125Z`	observation	∼22:00 MDT 07-16: spectra clean. (fieldnotes:p180)
07-17 09:45	`corr_20260717_154619Z`	photo	Three parallax photos of antenna from campsite (∼09:45 MDT). (fieldnotes:p180)
07-17 12:44	`corr_20260717_184434Z`	observation	Tx on/off test (∼12:44 MDT). (fieldnotes:p181)
07-17 12:50	`corr_20260717_185100Z`	mechanical	Lifted to final height 91 m (∼12:50 MDT). (fieldnotes:p181)
07-17 14:26	`corr_20260717_202615Z`	observation	Beam scan starts (∼14:26 MDT) after toggling free on/off. (fieldnotes:p181)
07-17 14:40–15:20	`corr_20260717_204117Z`	photo	Aaron hikes W slope N of cabin taking positioning photos. (fieldnotes:p181)
07-17 15:44–17:44	`corr_20260717_214419Z`	outage	Observing-daemon outage of ∼2 h in the middle of the beam scan. (derived)
07-17 16:00	`corr_20260717_220130Z`	power	7U sky battery failed holding [?] (∼16:00 MDT). (fieldnotes:p181)
07-17 17:35	`corr_20260717_233559Z`	outage	Phase C box-air (input 4) outage. (readme)
07-17 17:50	`corr_20260717_235101Z`	config-change	Peltier-support battery (∼17:50 MDT); LIDAR hooked up, raised for lift; normal before raising. (fieldnotes:p181)
07-17 19:37	`corr_20260718_013837Z`	observation	LIDAR sweep started in 5° increments (∼19:37 MDT 07-17). (fieldnotes:p181)
07-17 19:53	`corr_20260718_015339Z`	observation	Beam map resumed at az ∼75° and EL 180–180°. (fieldnotes:p181)
07-17 19:55	`corr_20260718_015548Z`	mechanical	EL lost motion facing down; stuck. (fieldnotes:p181)
07-17 21:02–21:22	`corr_20260718_030222Z`	data-anomaly	CORRECTED 2026-09-12. (derived)
07-17 21:05	`corr_20260718_032003Z`†	mechanical	CATASTROPHE (∼21:05 MDT 07-17): lost bottom pulley plate; seen splitting apart. (fieldnotes:p181)
07-17 21:08	`corr_20260718_032003Z`†	data-end	CORRECTED 2026-09-12. (derived)
07-17 21:24	—	data-gap	Corrupt source file skipped by filter (would have been the file after the last saved one). (readme)

B Masks

Generated from the mask catalog in `eigsep_data.select_files`. “Default” masks are applied unless disabled; others are opt-in.

mask	default	inputs	windows (UTC) and reason
snap-test-flip	always	all	07-12 21:15; 07-13 22:36; 07-14 02:07; 07-14 21:22; 07-15 00:25; 07-15 04:48; 07-16 01:22; 07-17 03:24. SNAP swapped to scratch board C000069 (not a flight configuration)
antenna-move-allzero	always	all	07-15 00:32–01:16. Phase B→C antenna move; spectra are identically zero
box-air-outage	yes	4, 5	07-15 18:55–21:30; 07-16 17:32–17:42; 07-16 18:42–23:27; 07-17 23:35–23:45. box-air (input 4) outage
phaseB-mux-off	yes	5	07-14 21:22–23:31. Phase B mux copy 4→5 disabled; the 35 cross is invalid
phaseC-mux-absent	yes	1, 5	07-15 01:17–03:29. Phase C mux copies not yet online; keys 1/5/15 unpopulated (inputs 0 and 4 themselves are valid here)
mux-tuning	yes	all	07-15 05:39–06:14; 07-15 06:47–07:16; 07-15 08:04–08:13; 07-15 08:51–10:53; 07-15 11:48–12:18; 07-15 13:34–14:50; 07-15 15:25–15:56. ADC mux configuration actively being iterated
boxair-emi-1mhz	yes	4, 5	07-16 01:18–16:51. box-air’s own 1.000 MHz comb EMI (teeth at integer MHz, ∼10 dB, walks across the channel grid; box-air only, 07-16 01:16:57-16:49:03, gone after the Panda power cycle).
thunderstorm	no	all	07-16 00:00–06:00. Thunderstorm with lightning visible in the data (end time estimated; field notes give onset only)
laptop-rfi	no	all	07-13 00:00–23:59. Laptop RFI 145-160 MHz with 2 MHz spacing during Phase A checks
end-of-campaign-writestall	no	all	07-18 03:20–03:22. Last 3 files flush-written 838-931 s late after a 1,061 s write stall; filenames lag their data badly.
accumulator-overflow	no	all	(per channel). int32 accumulator wrapped in at least one channel (negative auto power).
bad-header-clock	no	all	07-13 05:05–15:57; 07-12 13:31–20:39; 07-12 22:55–23:05; 07-15 03:19–03:28; 07-15 15:54–15:57. header/times unusable (unsynced F-engine); the DATA is fine.

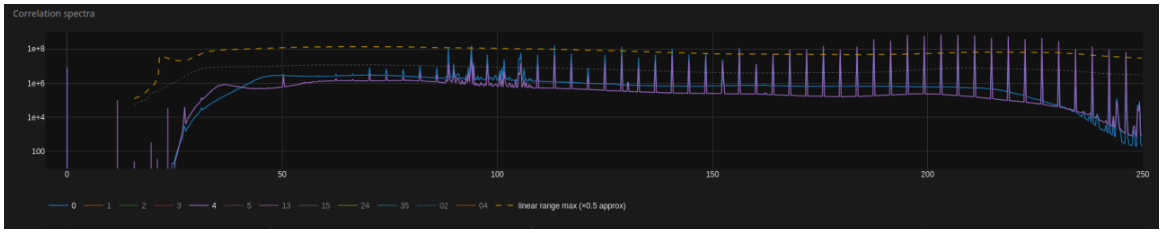
C Reproducing this memo

From this directory, with the `arp` environment:

```
python reduce_files.py      # per-file summary spectra -> cache/
python make_products.py     # figures/ and tables/
pdflatex memo-001.tex && pdflatex memo-001.tex
```

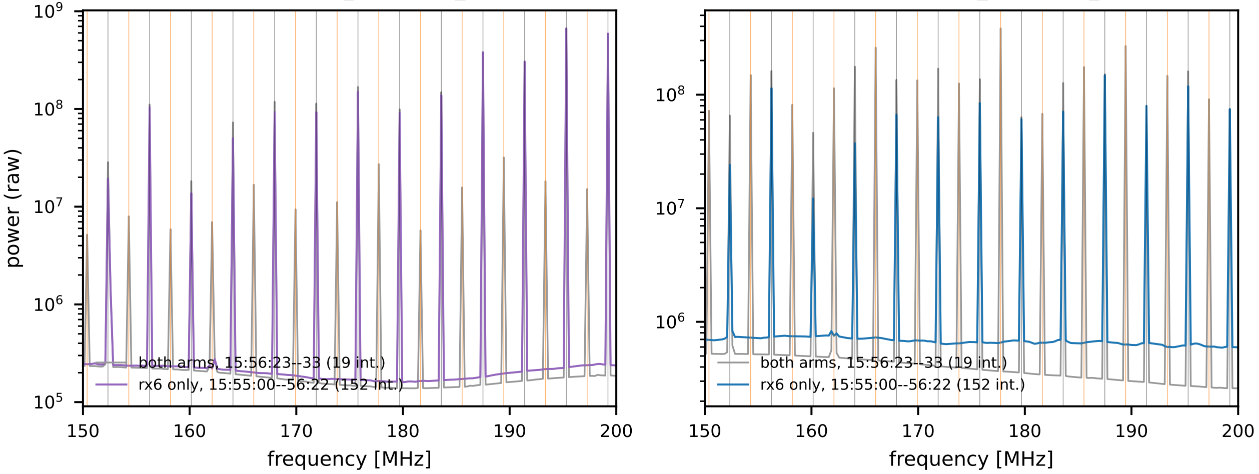
Inputs: `marjum-2026-07/data/*.h5`, `events.jsonl`, `curation/{mode_table,cal_windows}.jsonl`, `curation/pointing_table.parquet`, `imgs/fits/v0004_marjum_geometry/shared.json`, the mask catalog in `eigsep_data.select_files`, and `derived/dem/v0002/marjum_dem.npz` via `eigsep_terrain`. The anchor and tiedown GPS coordinates are in `make_products.py`.

repository	commit
eigsep (memo, campaign archive)	`00f468a`
eigsep_data (mask catalog)	`2b201d4`
eigsep_terrain (DEM access)	`6688c89`
DEM (`derived/dem/v0002`) SHA-256	`e890231c970be4ed`
geometry release v0004 `shared.json` SHA-256	`8fcf5360cbfa14a0`
LIDAR constraint v0002 `summary.json` SHA-256	`d684aa673f7c0d05`
transmitter feed v0002 `summary.json` SHA-256	`1af458e39f88f320`
generated	2026-10-10 05:18 UTC



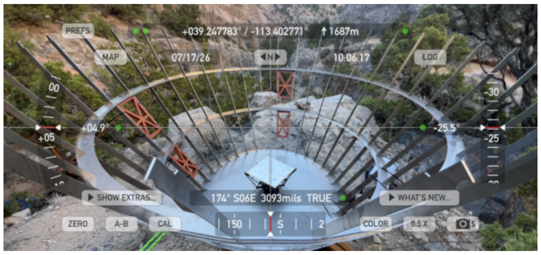
(a) live display, captured 07-17 15:56:29 UTC with only transmitter chain rx6 on (purple: input 4; blue: input 0)

Correlation spectra
1e+8
1e+6
1e+4
100
0
50
100
150
200
250
0
1
2
3
4
5
13
15
24
35
02
04
linear range max (×0.5 approx)



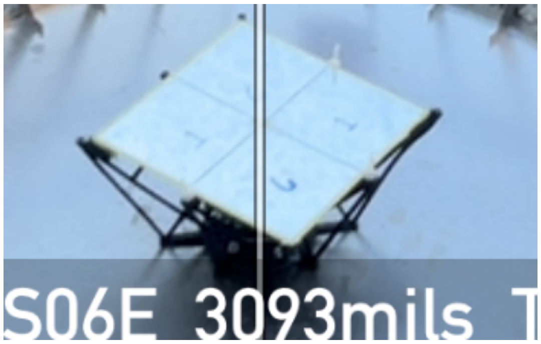
(b1) box-air (input 4), corr_20260717_155704Z.h5

(b2) box-gnd (input 0), corr_20260717_155704Z.h5



(c) IMG_2204: transmitter from above, heading 174 ° true

PREFS
+039.247783° / -113.402771°
↑1687m
MAP
07/17/26
◀N▶
10:06:17
LOG
00
+05
+04.9°
-25.5°
-30
-25
▶ SHOW EXTRAS...
174° S06E 3093mils TRUE
▶ WHAT'S NEW...
ZERO
A-B
CAL
150
S
2
COLOR
0.5 X
S



(d) feed: diagonal quadrants marked "6" = the rx6 dipole, roughly N--S

S06E 3093mils T

Figure 1: Which transmitter arm is which. (a) The live correlator display captured in the field with only transmitter chain rx6 powered (`imgs/tx_ant_rx6_on.png`). (b) The recorded spectra behind it: mean over the rx6-only integrations (colour) and over the next 10 s with both arms on (grey), 150–200 MHz; black lines mark channels ≡ 0 mod 16 (arm 0), orange ≡ 8 mod 16 (arm 1). With rx6 alone only the arm-0 teeth are present. Box-gnd's floor between the teeth is also about twice as high in the rx6-only interval. (c, d) IMG_2204 (`imgs/IMG_2204.PNG`), a theodolite-app capture looking down on the transmitter, and its feed, whose diagonal quadrants marked "6" form the rx6 dipole.

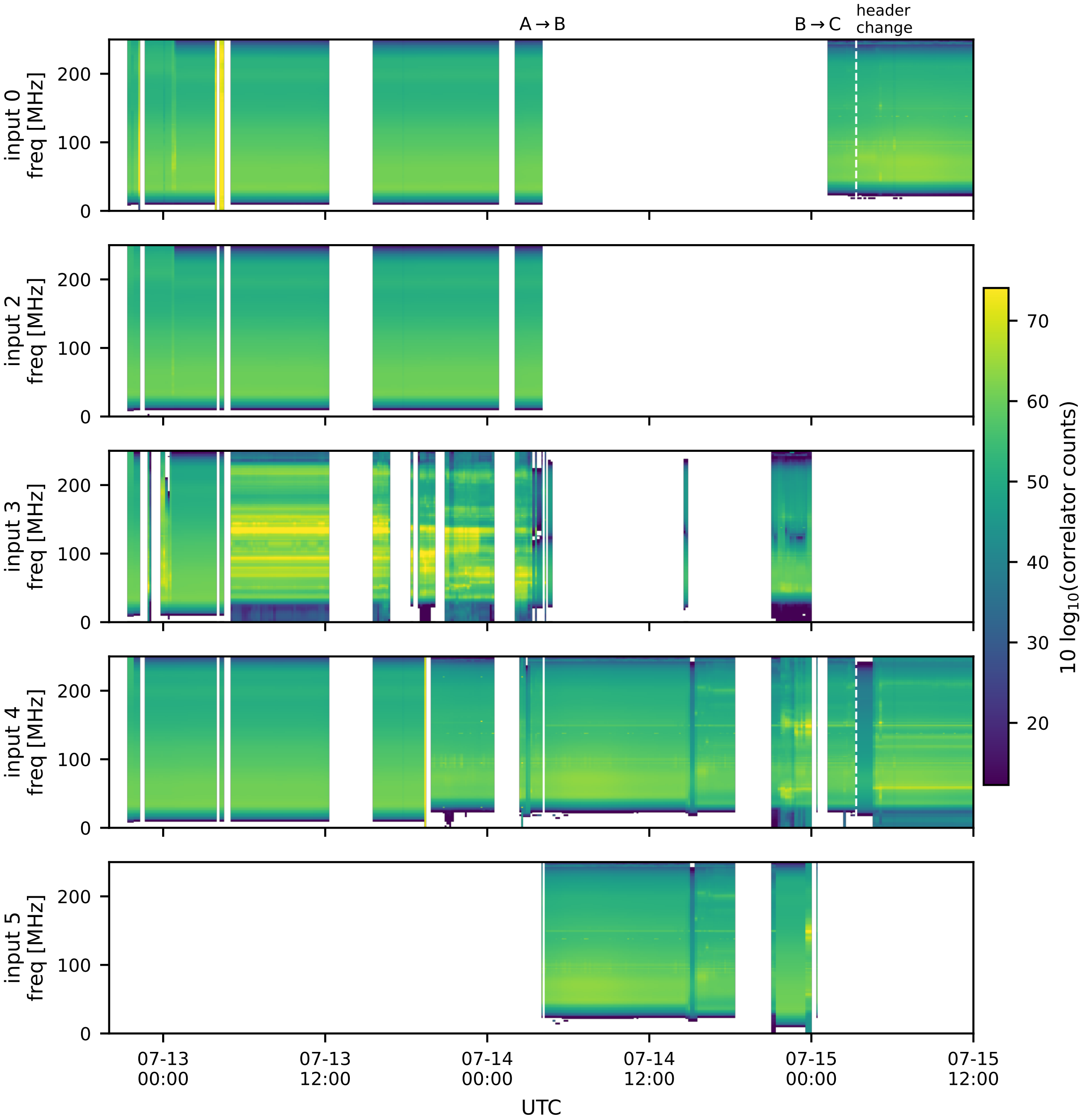


Figure 2: Power of every recorded input, 07-12 20:00 to 07-15 12:00: per-file median spectra in correlator counts, averaged into 10-minute and 2-MHz bins, on one colour scale. White is no data or an input the filter did not keep. Solid lines: wiring-phase changes found from the data; dashed: the change in the header `wiring`. Input 5 is the mux copy of input 4.

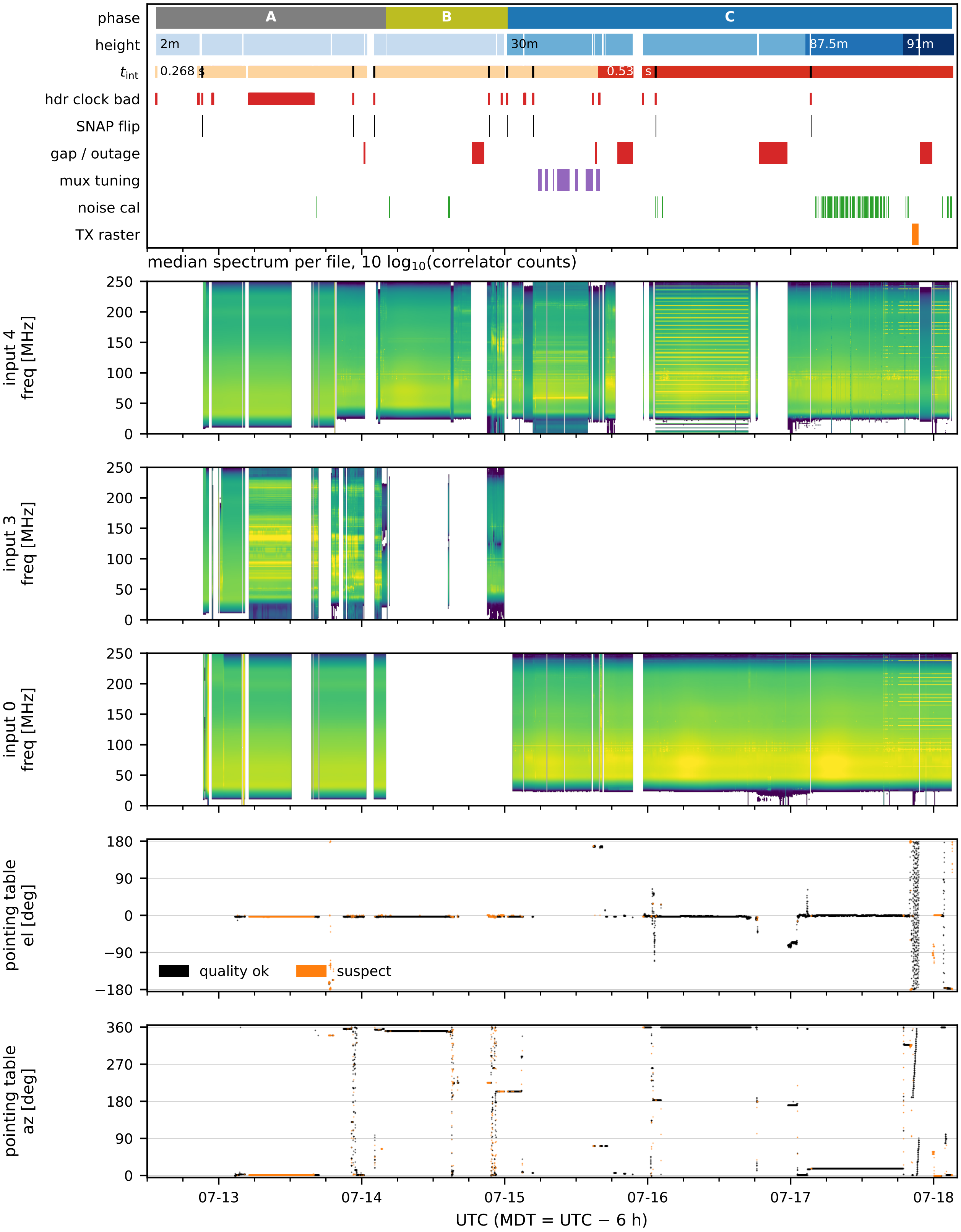


Figure 3: The campaign. *Top:* configuration and events: wiring phase; height era (from `curation/mode_table.jsonl`); integration time (ticks per file; black = the 4× SNAP test-flip files); files with a bad header clock; SNAP test flips; recording gaps and box-air outages; ADC-mux tuning; noise-source calibration windows; the transmitter raster. *Middle:* per-file median spectra of inputs 4, 3 and 0, as $10 \log_{10}$ of correlator counts (the 3 dB step at 07-15 15:55 is the integration doubling); white is no data or an unrecorded input. In phase C input 4 is box-air and input 0 is box-gnd. *Bottom:* elevation and (commanded) azimuth of box-air from `curation/pointing_table.parquet`, every 20th sample, coloured by the table's **`quality`**; conventions in §7.

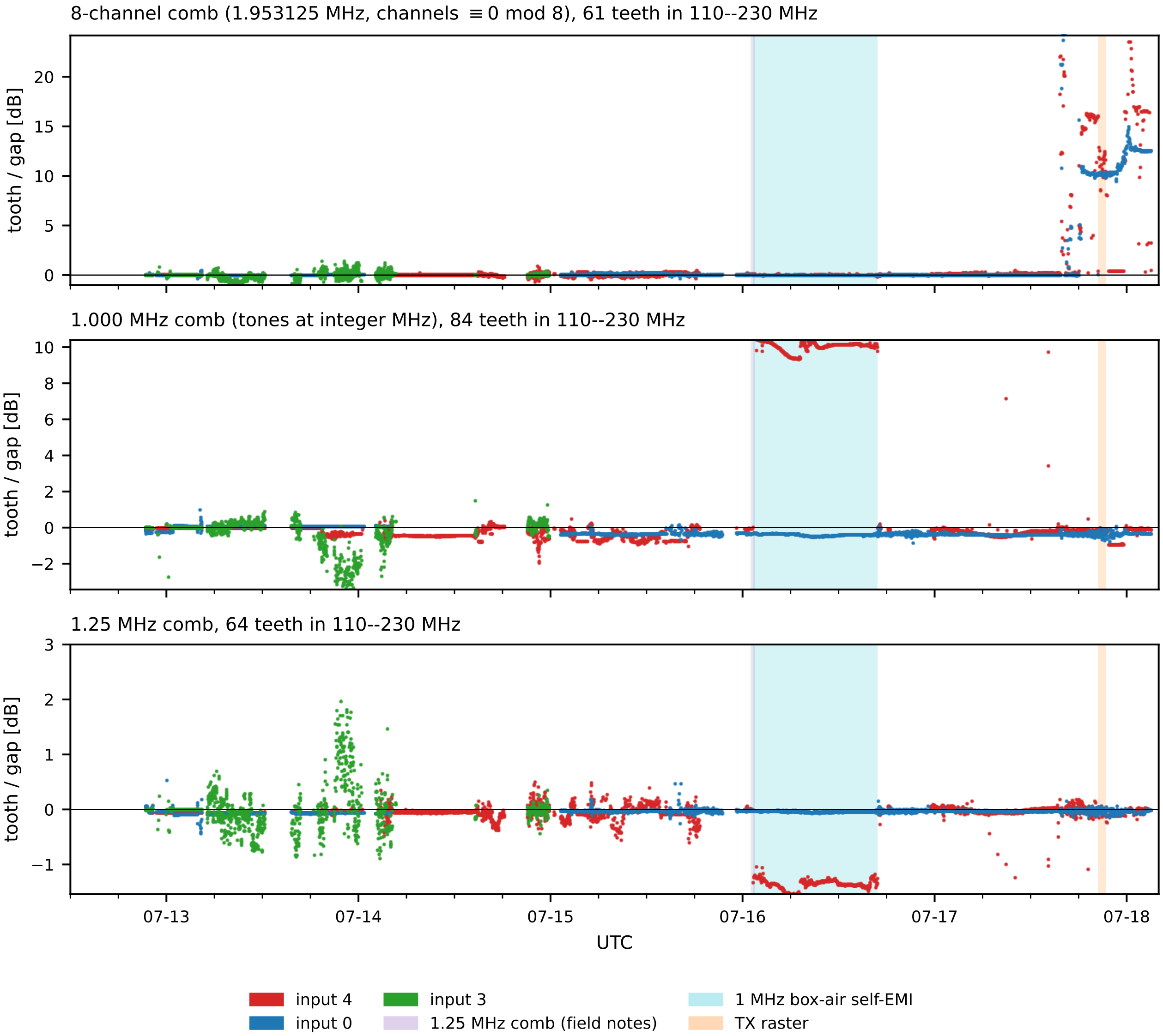


Figure 4: Comb contrast per file for inputs 4 (box-air in phase C), 0 (box-gnd in phase C) and 3 (phases A/B). Shaded: the field-note comb EMI window, the 1.000 MHz box-air self-EMI, and the 07-17 transmitter raster. The negative 1.25 MHz excursion on 07-16 is the 1.000 MHz comb landing in the 1.25 MHz gap channels, not a separate comb.

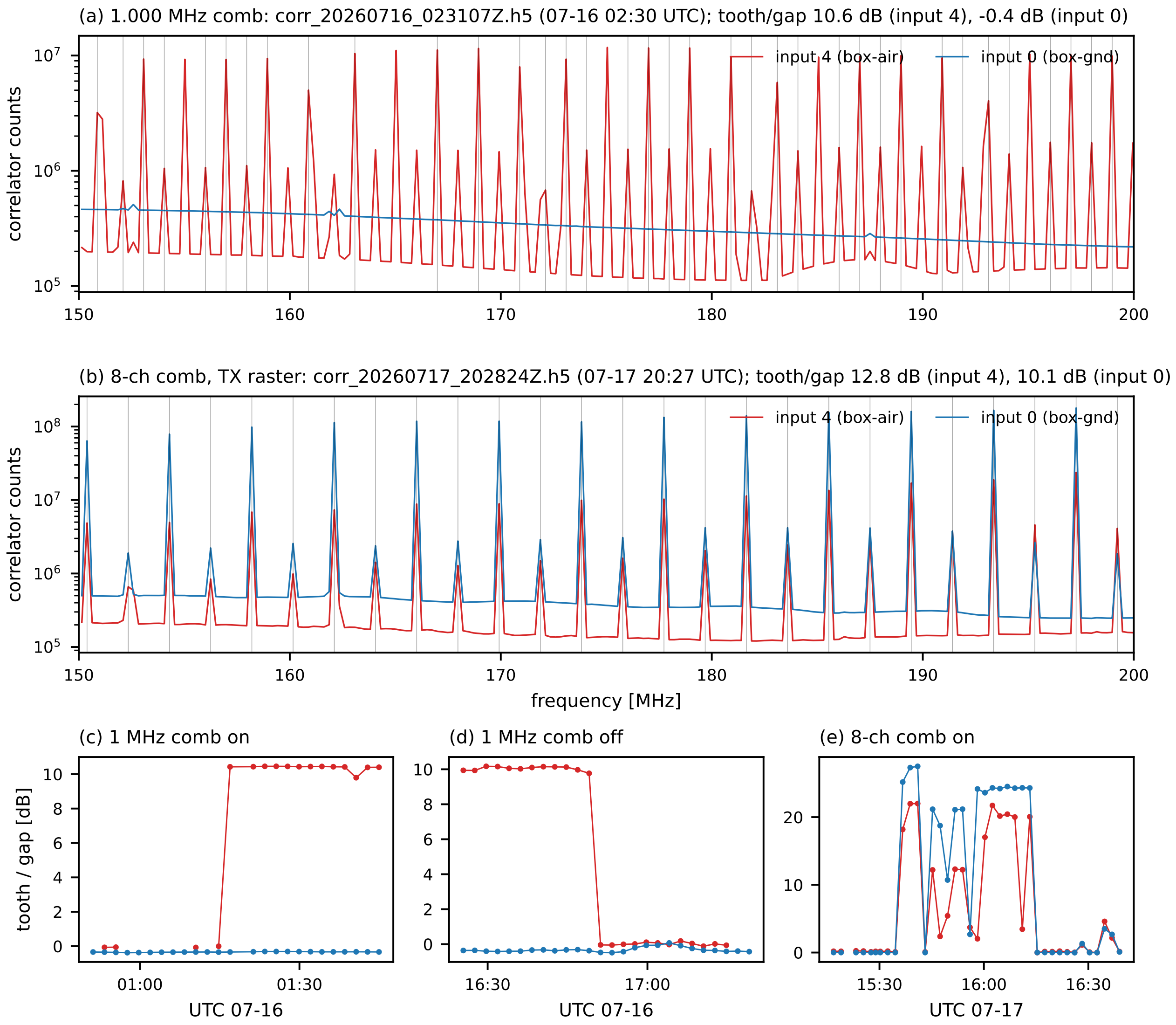


Figure 5: (a, b) Median spectra of the file with the strongest 1.000 MHz comb and of the file with the strongest 8-channel comb inside the transmitter raster, for box-air (red) and box-gnd (blue); grey lines mark the expected teeth. (c–e) Contrast per file around the on/off transitions.

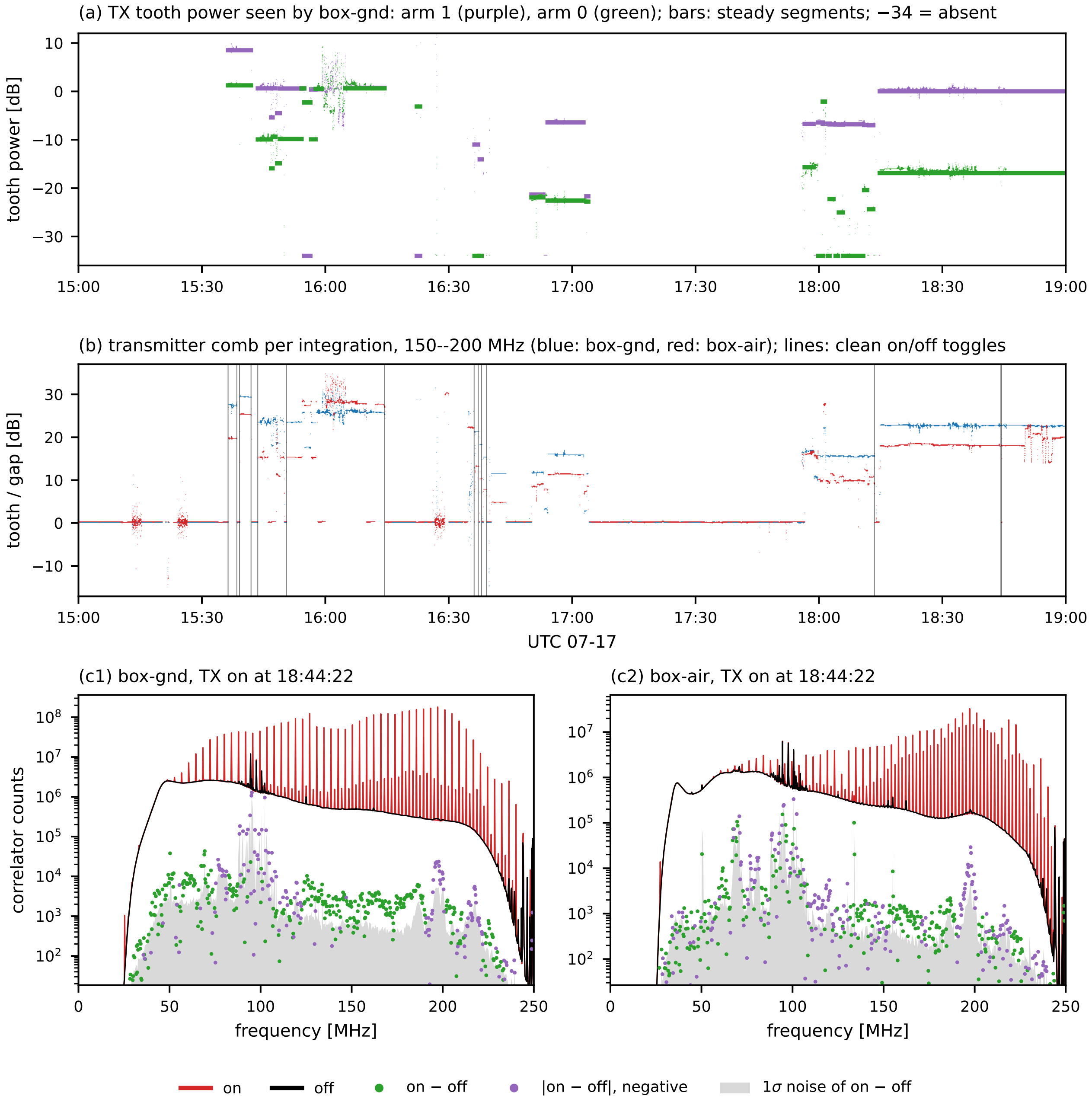
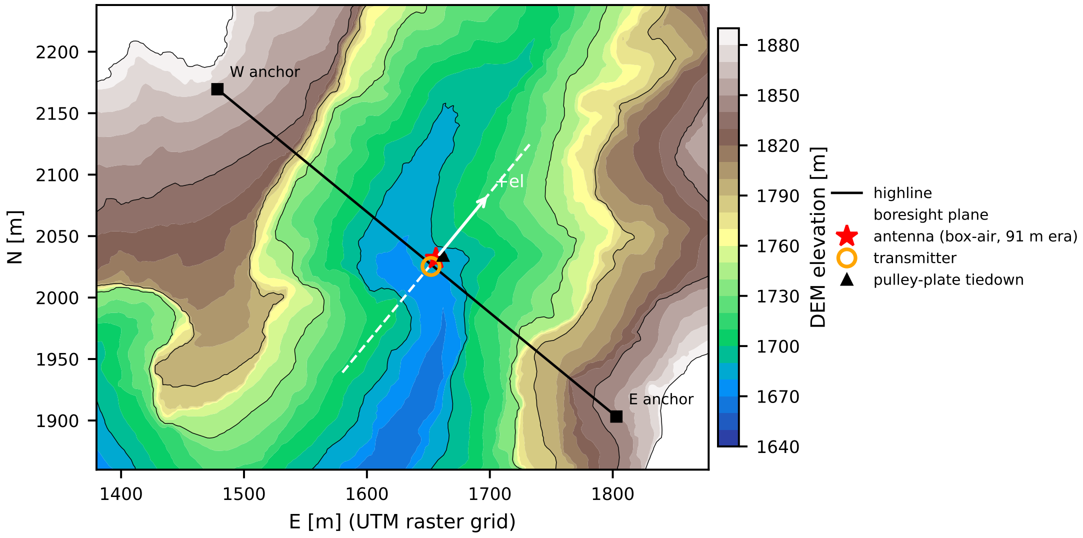
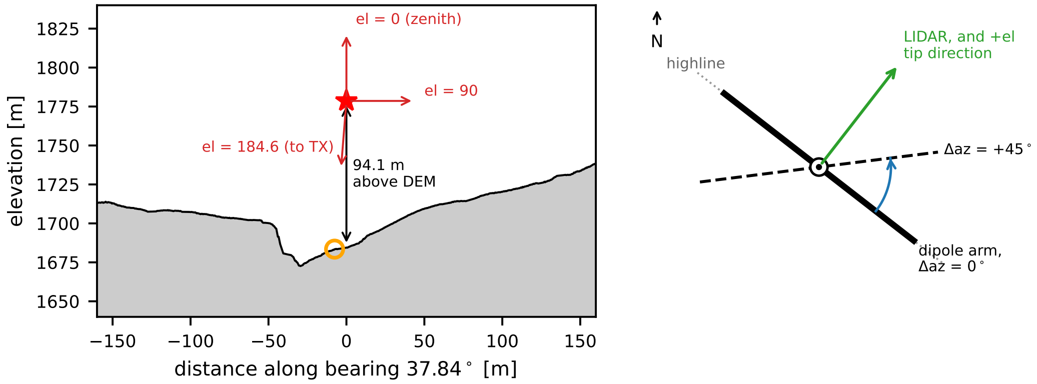


Figure 6: The 07-17 transmitter tests. (a) Tooth power of each transmitter arm seen by box-gnd, per integration, with the steady segments of Table 5. (b) Tooth/gap contrast per 0.54 s integration on both antennas; vertical lines mark the toggles used. (c) The last switch-on, in the final configuration: mean spectrum over the 14 integrations with the transmitter off, over the 14 with it on (skipping the first, which may be only partly on): on (red), off (black), on − off where positive (green), the magnitude of negative differences (purple dots), and the 1σ noise of the difference (grey shading, interpolated across the teeth). The difference, its negatives and the noise are shown only for channels at least two from any tooth (channel mod 8 in 2–6). Powers in correlator counts.



(a) plan view

W anchor
E anchor
+el
N [m]
2200
2150
2100
2050
2000
1950
1900
1400
1500
1600
1700
1800
E [m] (UTM raster grid)
DEM elevation [m]
1880
1850
1820
1790
1760
1730
1700
1670
1640
highline
boresight plane
antenna (box-air, 91 m era)
transmitter
pulley-plate tiedown



(b) section in the boresight plane

(c) Δaz, from above at el = 0, boresight out of page

Figure 7: (a) Plan view on the DEM, on the UTM raster grid of DEM `v0002` (EPSG:6341; true north lies 1.53° clockwise of grid north here). (b) Section through the antenna in the boresight plane, with the elevation convention. (c) The azimuth rotation seen from above at $\mathrm{el} = 0$, as $\Delta\mathrm{az}$, the rotation from the reference position in which the dipole arm lies along the highline ($\Delta\mathrm{az} = 0$): increasing $\Delta\mathrm{az}$ rolls the dipole about the boresight counter-clockwise seen from above.